Optimising Reviewer Assignments with Max-Flow

A C++ Implementation of the Edmonds-Karp Algorithm

Made with GAMMA

Project Overview

Automating the matching of scientific paper submissions to reviewers by treating the assignment process as a Network Flow problem.

Ensures every paper meets required review counts whilst respecting each reviewer's maximum workload limits.

Made with GAMMA

Core Data Architecture

Entities

Submission and Reviewer structs store IDs, emails, and expertise levels (Primary and Secondary).

ProjectConfig

Centralised structure managing parameters like minReviewsPerSubmission and maxReviewsPerReviewer.

Data Mapping

Uses std::map to translate abstract Graph Node IDs back into real-world Database IDs.

Made with GAMMA

The Flow Network Model

The createGraph class transforms raw CSV data into a directed flow network with four distinct layers:

Super Source (S)

Distributes flow to all Submissions

Submissions Layer

Nodes representing papers with capacity set by "Min Reviews Needed"

Reviewers Layer

Nodes representing experts with capacity set by "Max Workload"

Super Sink (T)

Collects the final flow to complete the assignment

Made with GAMMA

Algorithmic Core: Edmonds-Karp

BFS Search

Uses Breadth-First Search to find the shortest augmenting paths in the residual graph.

Bottleneck Analysis

Identifies the minimum available capacity along a path to push flow.

Residual Edges

Implements reverse edges to allow the algorithm to "undo" and optimise assignments during execution.

Made with GAMMA

Assignment Logic & Matching Modes

Flexible matching based on configuration settings:

1

Primary-Primary Only

Matches Primary Submission domains to Primary Reviewer expertise only.

2

Primary-Flexible

Allows Primary Reviewers to take on both Primary and Secondary Submission domains.

3

Full Exhaustive

Exhaustive matching considering all combinations of Primary/Secondary expertise for both parties.

Made with GAMMA

Interactive & Batch Execution

Interactive Menu

User-friendly CLI to load datasets, build the network, and step through the Edmonds-Karp execution.

Batch Mode

Command-line flag (-b) for rapid processing of large-scale datasets, exporting results directly to CSV.

Made with GAMMA

Results & Risk Analysis

Assignment CSV

Detailed mapping of every Paper-to-Reviewer match.

Missing Reviews

Specifically flags "Risky" submissions that could not be fully assigned due to expertise gaps or reviewer shortages.

Path Visualisation

Displays the exact flow path (Source → Sub → Rev → Sink) for every successful match.

Made with GAMMA

Conclusion

By modelling assignments as a flow problem, we guarantee a mathematically optimal distribution of work.

This C++ solution ensures fairness for reviewers and high-quality matches for conference organisers.

Made with GAMMA